Database Design Exercise 5

姓名：易治行　学号：2023141220023

I Automobile Company Database

Problem statement

Design a database for an automobile company to provide to its dealers to assist them in maintaining customer records and dealer inventory and to assist sales staff in ordering cars.

Each vehicle is identified by a vehicle identification number (VIN). Each individual vehicle is a particular model of a particular brand offered by the company. Each model can be offered with a variety of options, but an individual car may have only some or none of the available options. The database needs to store information about models, brands, options, dealers, customers, and cars.

The design includes an ER diagram, relation schemas, and primary-key, foreign-key, and semantic constraints.

Answer

1 ER Diagram

The ER model separates available options for a model from options actually installed on a vehicle. It also connects dealer inventory and customer orders to individual vehicles.

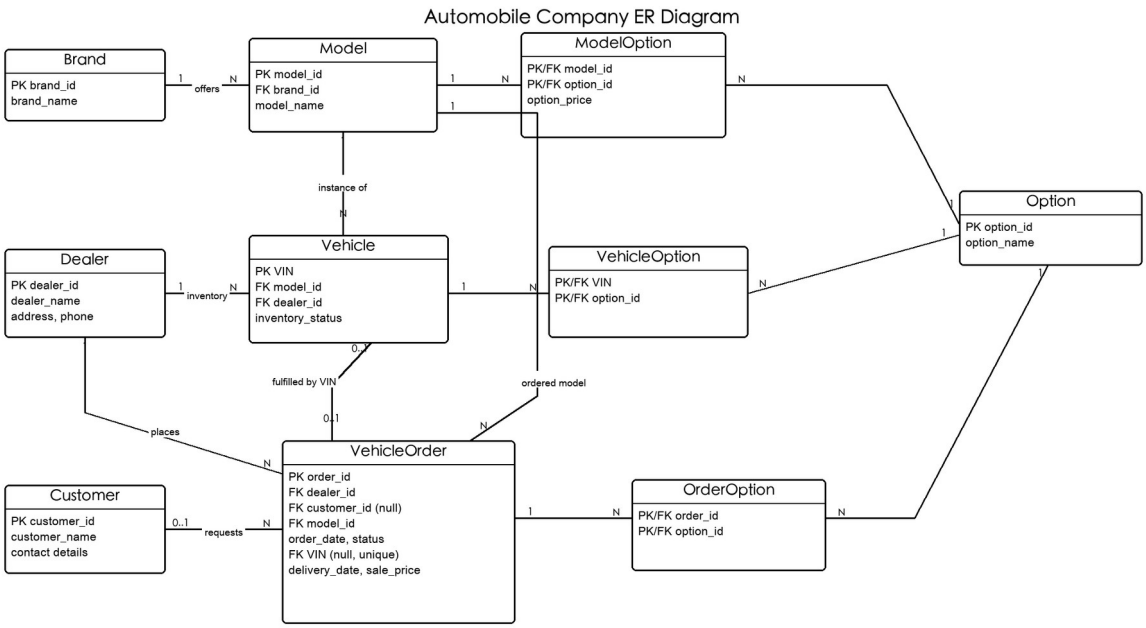


Figure 1　Automobile company ER diagram

2 Relation Schemas

BRAND(brand_id, brand_name)
PK: brand_id　UNIQUE: brand_name

MODEL(model_id, brand_id, model_name)
PK: model_id FK: brand_id -> BRAND(brand_id)
UNIQUE: (brand_id, model_name)

OPTION(option_id, option_name, description)
PK: option_id UNIQUE: option_name

MODEL_OPTION(model_id, option_id, option_price)
PK: (model_id, option_id)
FK: model_id -> MODEL(model_id)
FK: option_id -> OPTION(option_id)

DEALER(dealer_id, dealer_name, address, phone)
PK: dealer_id

CUSTOMER(customer_id, customer_name, phone, email, address)
PK: customer_id

VEHICLE(VIN, model_id, dealer_id, inventory_status)
PK: VIN
FK: model_id -> MODEL(model_id)
FK: dealer_id -> DEALER(dealer_id)

VEHICLE_OPTION(VIN, option_id)
PK: (VIN, option_id)
FK: VIN -> VEHICLE(VIN)
FK: option_id -> OPTION(option_id)

VEHICLE_ORDER(order_id, dealer_id, customer_id, model_id, order_date,
order_status, VIN, delivery_date, sale_price)
PK: order_id
FK: dealer_id -> DEALER(dealer_id)
FK: customer_id -> CUSTOMER(customer_id), nullable
FK: model_id -> MODEL(model_id)
FK: VIN -> VEHICLE(VIN), nullable and UNIQUE

ORDER_OPTION(order_id, option_id)
PK: (order_id, option_id)
FK: order_id -> VEHICLE_ORDER(order_id)
FK: option_id -> OPTION(option_id)

3 Constraints

1. VIN and all primary-key values are unique and not null.
2. Each model belongs to exactly one brand; model names are unique within a brand.
3. option_price and sale_price must be greater than or equal to zero.
4. inventory_status is restricted to ORDERED, IN_STOCK, RESERVED, or SOLD.
5. order_status is restricted to PLACED, CONFIRMED, BUILT, DELIVERED, or CANCELLED.
6. A VEHICLE_OPTION must be listed in MODEL_OPTION for that vehicle's model.
7. An ORDER_OPTION must be listed in MODEL_OPTION for the order's model.
8. A fulfillment VIN must refer to the same model and dealer as the order.
9. A delivered customer order has non-null VIN, delivery_date, and sale_price; delivery_date is not earlier than order_date, and the vehicle status is SOLD.

II 医院数据库系统

题目

请为某医院设计数据库系统，需要管理以下信息：该医院下设若干科室（科室号，科室名），每个科室有若干医生（职工号，姓名，性别，职称）和护士（职工号，姓名，性别），每位医生和护士都只在一个科室工作。病人（身份证号，姓名，性别，出生日期）在科室住院，一位病人可能多次在同一科室住院，因此需要记录每次住院的入院日期和出院日期。

1. 请分析上述描述，并充分利用常识，画出该系统的 ER 图；

2. 将你设计的 ER 图转化为一组合适的关系模式，需给出主码、外码约束。

答案

1 ER 图

科室与职工之间是一对多关系，每位职工只属于一个科室。医生和护士是职工的互斥且完全的子类型。病人与科室之间的多对多住院关系由“住院记录”实体表示，入院日期用于区分同一病人在同一科室的多次住院。

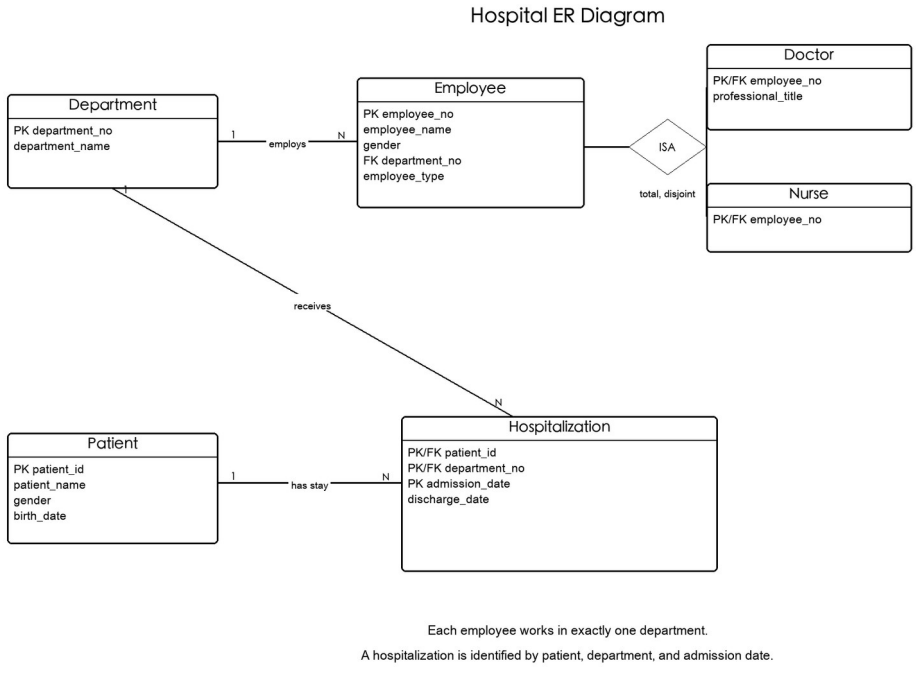


Figure 2　Hospital ER diagram

2 关系模式

DEPARTMENT(department_no, department_name)
PK: department_no　UNIQUE: department_name

EMPLOYEE(employee_no, employee_name, gender, department_no, employee_type)
PK: employee_no
FK: department_no -> DEPARTMENT(department_no)

DOCTOR(employee_no, professional_title)
PK: employee_no
FK: employee_no -> EMPLOYEE(employee_no)

NURSE(employee_no)
PK: employee_no
FK: employee_no -> EMPLOYEE(employee_no)

PATIENT(patient_id, patient_name, gender, birth_date)
PK: patient_id

HOSPITALIZATION(patient_id, department_no, admission_date, discharge_date)
PK: (patient_id, department_no, admission_date)
FK: patient_id -> PATIENT(patient_id)
FK: department_no -> DEPARTMENT(department_no)

3 其他约束

1. 每位职工必须且只能属于一个科室，department_no 不允许为空。

2. employee_type 只能取 DOCTOR 或 NURSE。

3. 职工的专门化是完全且互斥的：每个 EMPLOYEE 必须且只能出现在 DOCTOR、NURSE 中的一个表，并与 employee_type 一致。

4. 性别属性只能取医院规定的合法值。

5. 当前仍在住院时，discharge_date 可以为空；否则必须满足 discharge_date >= admission_date。

6. 同一病人的两段住院时间不能重叠。

7. 复合主码 (patient_id, department_no, admission_date) 允许同一病人在不同时间多次入住同一科室。